Mohamed Mostafa Metawea

AI Engineer · Generative AI & Agentic Systems

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PROFESSIONAL SUMMARY

Motivated AI and Machine Learning engineer with a strong foundation in machine learning, deep learning, and generative AI. Experienced in building end-to-end projects, including data processing, model development, and deployment using modern tools and frameworks. Skilled in Python, NLP, and computer vision, with growing expertise in LLM applications and scalable AI systems. Eager to apply technical knowledge, learn quickly, and contribute to impactful real-world projects in a collaborative environment.

TECHNICAL SKILLS

• **Programming:** Python, SQL, C++, API Development, Object-Oriented Programming

• **Machine Learning:** Regression, Classification, Clustering, SVM, KNN, Decision Trees, Random Forest, Ensemble Methods, Feature Engineering, Model Evaluation

• **Deep Learning:** Neural Networks (ANN, DNN), CNN, RNN, LSTM, GRU, Transfer Learning, Attention Mechanisms, Transformers

• **Generative AI & LLMs:** Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, LLM Applications, OpenAI API, Embeddings, Vector Databases, Multimodal AI, Fine-tuning (LoRA, PEFT)

• **Computer Vision:** Image Analysis, CNN-based Vision Models, Vision Transformers (ViT), Transfer Learning

• **NLP:** Tokenization, Text Preprocessing, TF-IDF, Word2Vec, GloVe, N-grams, POS Tagging, Named Entity Recognition (NER), Text Classification, Sentiment Analysis, Summarization, Question Answering

• **Frameworks & Libraries:** TensorFlow, Scikit-learn, NumPy, Pandas, Hugging Face, LangChain, Gensim, FastText

• **Data & Databases:** MongoDB, Vector Databases, Data Pipelines, Data Preprocessing, Data Visualization, Power BI

• **MLOps & Deployment:** Docker, FastAPI, MLflow, Model Deployment, API Integration, CI/CD Concepts, Cloud Deployment, Monitoring & Debugging

• **Cloud & Tools:** Microsoft Azure (AI Engineer), AWS AI Fundamentals, Cloudflare Workers, Jupyter Notebook, Git, OpenAI, Model Context Protocol (MCP)

• **Automation**: n8n

PROFESSIONAL EXPERIENCE

Feb 2025 – Mar 2025 **Machine Learning Intern** | *Cellula Technologies*

• Developed machine learning models for predictive maintenance, improving model performance using cross-validation and feature engineering.
• Containerized and deployed 3 ML models as REST APIs using Flask and Docker, enabling real-time inference and consistent deployment environments.

AI & DATA SCIENCE PROJECTS

MedAgent — Autonomous Clinical Orchestrator

DEPI Graduation Project · LangGraph · GPT-4o · RAG · FAISS · FastAPI · Docker · AES-256 · Prometheus · OpenTelemetry

• Built a production-grade multi-agent AI system with 14 specialized agents for clinical triage, diagnosis, and decision support using LangGraph and GPT-4o.
• Implemented Retrieval-Augmented Generation (RAG) using FAISS over medical guidelines (NIH, WHO), reducing hallucinations and improving response reliability.
• Designed a Tree-of-Thought reasoning pipeline with fallback strategies, enabling multi-step decision-making with confidence scoring.
• Secured the system with AES-256 encryption, JWT-based RBAC, and audit logging using SHA-256 hash chains.
• Deployed using FastAPI, Docker, and Streamlit with full observability via Prometheus and OpenTelemetry.

AI Personal Assistant — Multi-Agent System | n8n, OpenRouter, Groq, Gemini *Personal Project*

- Built a production-ready multi-agent AI system with a central Orchestrator routing tasks across 9 specialized sub-agents (Calendar, Email, Search, Weather, Notes, Reminders, Social Media, Data Analysis, Web Automation) via WhatsApp & Telegram

- Implemented multi-modal input (voice via Groq Whisper, images via Gemini 2.0 Flash, PDF extraction) and reduced operational cost from ~$120/month to $0 using free-tier APIs
- Integrated Google Calendar, Gmail, Google Sheets, LinkedIn, and Facebook APIs with OAuth2 authentication and per-user session memory

AI-Based Face Recognition Attendance System | *Python · CNN · OpenCV · Flask · Liveness Detection*

• Developed a CNN-based face recognition system achieving 95% accuracy with sub-second inference latency.
• Implemented liveness detection and anti-spoofing, preventing 98% of replay attacks.
• Automated attendance workflow, reducing manual effort by 80%.

Heart Disease Risk Prediction System | *LightGBM · XGBoost · SHAP · Streamlit*

- Built a LightGBM/XGBoost ensemble achieving 92% recall and 0.90 F1-score on held-out clinical data, optimized for high sensitivity to minimize false negatives; applied SHAP to identify top risk predictors and make the model auditable for clinical review. Deployed as an interactive Streamlit web app.

Multilabel Comment Toxicity Detection | *NLP · TF-IDF · Scikit-learn · Threshold Optimization*

- Developed a multi-label NLP classification model using TF-IDF and Scikit-learn
- Reduced false positives by 18% using threshold optimization and feature engineering.

EDUCATION

2020 – 2025 **B.Sc. Communication & Electronics Engineering** | *October 6 University, Giza, Egypt*

- GPA: 3.44 / 4.0 · Graduation Project: **AI-Based Face Recognition Attendance System** (Grade A)

CERTIFICATIONS & COURSES

*** Deep Learning Specialization** — DeepLearning.AI / Coursera *(Andrew Ng)*
*** IBM AI Engineering Professional Certificate** __ Coursera
*** Generative AI Professional —** Digital Egypt Pioneers Initiative (DEPI), Ministry of Communications & Information Technology **(Nov 2025 – Jun 2026)**
*** AI Engineering Specialization (9 Courses)** — Scrimba / Coursera *(2025–2026)*
*** Post-Training Large Language Models: Fine-Tuning & RLHF** — DeepLearning.AI *(Andrew Ng & Sharon Zhou) (2025–2026)*
*** AI Training Program** — National Telecommunication Institute (NTI), Egypt *(Oct 2025 – Jan 2026)*
*** Machine Learning Specialization** — DeepLearning.AI / Coursera *(Andrew Ng)*
*** IBM Data Science Professional Program** — Digital Egypt Pioneers Initiative (DEPI), Ministry of Communications & Information Technology *(Oct 2024 – Jun 2025)*
*** Orange AI Level 1 Certification** — Orange Digital Center *(Nov 2024)*

LANGUAGES

- Arabic — Native · English — Professional Working Proficiency